

SonicSentinel AI

Project documentation

Evidence-led technical showcase

Audio intelligence with a reviewable path

Product

React/Vite frontend, FastAPI service, persistence, and ML boundary

Operational audio is difficult to triage at scale

Context

Detection needs confidence, severity, auditability, and human review

Detect, compare, alert, and document

Objectives

Upload/live capture, model comparison, alerts, review workflow

A dual-engine monitoring workflow

Solution

Input -> validation -> inference -> confidence -> alert/review

Separated application boundaries

Verified source

Frontend -> FastAPI -> SQLite/uploads; model boundary is explicit

The inspected implementation surface

Stack

React, Vite, TypeScript, Framer Motion, Recharts, FastAPI, Python, SQLite

SRS mapping is part of the deliverable

Evidence

See SRS_COMPLIANCE_MATRIX.md; statuses are mapped to inspected evidence

Runtime audio inventory

Observed

WAV upload inventory generated from ml-service/data/uploads

Labels are not present in upload filenames

Status

Class distribution: PENDING VERIFICATION

Readable WAV metadata was inventoried

Evidence

See [dataset/audio-inventory.csv](#) and [quality-report.md](#)

Input validation precedes inference

Pipeline

Container validation and feature extraction implementation require runtime evidence

Feature provenance must be reproducible

Status

Exact training features and parameters: PENDING VERIFICATION

Model evidence boundary

Status

RF metrics/artifact: RESULTS PENDING

Model evidence boundary

Status

SVM metrics/artifact: RESULTS PENDING

Model evidence boundary

Status

CNN metrics/artifact: RESULTS PENDING

Dataset to deployable artifact

Status

Training run artifacts are not present in the repository

Evidence before numbers

Status

No accuracy, precision, recall, F1, or confusion matrix is invented

Class-level diagnostics

Status

RF/SVM/CNN matrices: RESULTS PENDING

Agreement is a workflow decision

Status

Comparison CSV is explicitly marked RESULTS PENDING

Confidence should route uncertainty

Implementation

Detections retain prediction data through API routes; saved samples pending

Severity becomes an operator action

Routes

Alerts exposed by GET /alerts and PATCH /alerts/{alert_id}

Frontend route and screen inventory

UI

Landing, auth, dashboard, analyze, live, history, alerts, reports, models, settings

Persistence boundary

Observed

SQLite files and entities are implemented in `ml-service/app/sqlite_db.py`

FastAPI route catalog

Verified

Full method/path list is in API_DOCUMENTATION.md

Session-aware protected access

Routes

Register, login, me, profile, logout, reset routes are present

Keep secrets and microphone consent server-bound

Status

Security test evidence: PENDING VERIFICATION

Build evidence exists; functional evidence remains

Verified

npm run build passed; test suite artifacts are pending

Traceability from requirement to evidence

Matrix

See SRS_COMPLIANCE_MATRIX.md

Separate observed facts from open claims

Summary

22 diagrams created; model metrics and screenshots pending

What this package does not claim

Honesty

No fabricated model numbers, screenshots, or runtime test outcomes

Close the evidence gaps

Next steps

Capture browser screens, run ML evaluation, add labelled dataset manifest

A credible technical record

Close

Every open claim has an explicit evidence status and file location